AR-15 / M4 / M16 Platform

Colt / Aero Precision / BCM / PSA / US Military • Direct Gas Impingement Carbines & Rifles (5.56x45mm NATO & .223 Rem / 300 BLK)

Manufacturer:	Colt / Aero Precision / BCM / PSA / US Military	**Category:**	Rifles & Carbines
Action Type:	Direct Gas Impingement Semi-Automatic	**Era / Service:**	American Standard Service Rifle (1962–Present)
Calibers:	5.56x45mm NATO, .223 Remington, 300 AAC Blackout, 6.5 Grendel	**Covered Models:**	M4 / M4A1 Carbine (14.5"), M16A2 / M16A4 (20"), AR-15 Carbine (16"), Mk18 CQBR (10.3"), Mk12 SPR

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Gas Key Screws MO Staking (35–40 in-lbs) • Barrel Nut Torque (30–80 ft-lbs) • Castle Nut Staking (38–42 ft-lbs) • Headspace Verification (GO / NO-GO / FIELD) • Gas Tube Port Alignment

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
BARREL NUT TORQUE	Initial torque 30 ft-lbs, then tighten to align gas tube notch; DO NOT exceed 80 ft-lbs (apply Aeroshell 64 / Mil-G-21164D grease to threads)
GAS KEY SCREWS	Torque to 35 to 40 in-lbs; stake carrier metal into screw head counterbores with dedicated MO carrier staking tool
CASTLE NUT TORQUE	38 to 42 ft-lbs; stake receiver end plate into castle nut notch in two locations
HEADSPACE 556 NATO	GO = 1.4636", NO-GO = 1.4670", FIELD = 1.4730" (measured with stripped bolt without ejector/extractor)
FIRING PIN PROTRUSION	0.028" minimum to 0.036" maximum through bolt face
BOLT TAIL CARBON	Clean carbon off bolt tail; excess carbon creates friction and short stroking
GAS RINGS	Three staggered split gas rings; bolt assembly must support carrier weight when stood vertically on bolt face (no collapsing under own weight)

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Short Stroking / Bolt Over Base Malfunction **Bolt carrier recoils partially, fails to strip fresh round or pinches case in ejection port (stovepipe)**	• Loose carrier gas key leaking gas (un-staked or broken gas key screws). • Misaligned gas block choking the barrel gas port. • Worn or broken bolt gas rings (rings aligned or failed stand test). • Excessive buffer weight (running H3 buffer on under-gassed carbine).	**Immediate Action: TAP magazine floorplate, RACK charging handle aggressively, ASSESS feedway. Gunsmith Action: Inspect carrier gas key. If loose, clean threads, apply Loctite 271, torque to 40 in-lbs and stake metal into screw notches. Check gas tube alignment into carrier key (tube must slide smoothly into key with zero resistance).**

Stoppage 2: Failure to Extract (Spent Casing Stuck in Chamber) **Bolt unlocks and travels rearward, but extractor rips off casing rim leaving fired case inside chamber**	• Fouled or corroded chamber creating severe casing drag during peak chamber pressure. • Weak extractor spring lacking 5-coil heavy-duty spring or black elastomer buffer insert. • Over-gassed carbine timing (bolt unlocking while residual pressure is still holding case expanded against chamber wall).	**Gunsmith Action: Use chamber brush with solvent on drill at slow speed to polish chamber. Detail strip bolt: replace extractor spring with Colt 5-coil gold spring, black rubber buffer, and viton O-ring. If over-gassed, install heavier H2 buffer or adjustable gas block.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
AR-15 / M4 Armorer Maintenance: Complete Disassembly	Brownells Gunsmithing Bench	Full armorer teardown of the AR-15 upper and lower receiver, bolt carrier group disassembly, gas rings inspection, and trigger group service.
The Brownells AR-15 Maintenance Guide: Inspection & Assembly	Brownells Gunsmithing Bench	Comprehensive armorer guide: Checking headspace with Clymer gauges, carrier key staking, barrel nut torque, and buffer tube assembly.